

Summer Bootcamp

Session 3 – Optimization

Optimality, convexity, algorithms, constraints, and finance

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Roadmap

- 1 Formulating an optimization problem
- 2 Unconstrained optimality
- 3 Convexity: from local to global
- 4 Algorithms for smooth optimization
- 5 Equality constraints and Lagrange multipliers
- 6 Inequality constraints and KKT conditions
- 7 Portfolio optimization
- 8 Putting the toolkit to work

Section 1

Formulating an optimization problem

The common language

A generic optimization problem is

$$\min_{x \in \mathbb{R}^d} f(x) \quad \text{subject to} \quad h(x) = 0, \quad g(x) \leq 0.$$

- x : decision variable;
- f : objective function;
- $\mathcal{F} = \{x : h(x) = 0, g(x) \leq 0\}$: feasible set;
- x^* : optimizer, if $f(x^*) \leq f(x)$ for every feasible x .

Maximizing $f(x)$ is the same as minimizing $-f(x)$.

Three recurring financial problems

Calibration / regression

$$\min_{\theta} \frac{1}{2} \|A\theta - b\|_2^2.$$

Calibration means choosing model parameters θ so that model outputs are as close as possible to observed market or historical data.

Minimum-variance portfolio

$$\min_w \frac{1}{2} w^T \Sigma w \quad \text{s.t.} \quad \mathbf{1}^T w = 1.$$

Constrained allocation

$$\min_w \frac{1}{2} w^T \Sigma w - \gamma \mu^T w \quad \text{s.t.} \quad \mathbf{1}^T w = 1, \quad w \geq 0.$$

Worked Example: Translate Words into Mathematics

A trader fits a linear model $y_i \approx a + bt_i$ and wants the smallest squared error.

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$$x = \begin{pmatrix} a \\ b \end{pmatrix}, \quad \min_{a,b} \frac{1}{2} \sum_{i=1}^n (a + bt_i - y_i)^2.$$

There is no constraint: $\mathcal{F} = \mathbb{R}^2$.

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In matrix form, with $A_i = (1, t_i)$ and $y = (y_i)$,

$$\min_x \frac{1}{2} \|Ax - y\|_2^2.$$

Your Turn: Formulate a Constrained Tracking Problem

A manager chooses weights $w \in \mathbb{R}^d$ so that portfolio returns $w^\top r_t$ track benchmark returns b_t , where b_t is the return at date t of a reference index or strategy used to evaluate the portfolio. The portfolio must be fully invested and long-only.

Task

Identify the decision variable, objective, and feasible set.

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Task

Identify the decision variable, objective, and feasible set.

With R containing the asset returns and b the benchmark returns,

$$\min_{w \in \mathbb{R}^d} \frac{1}{2} \|Rw - b\|_2^2 \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w \geq 0.$$

The decision variable is w and the feasible set is the long-only simplex.

Local versus global minimum

- x^* is a **local minimum** if it wins against nearby feasible points.
- x^* is a **global minimum** if it wins against all feasible points.

The central difficulty

Derivatives provide local information. Optimization usually asks for a global answer.

Convexity will be the bridge between the two.

Section 2

Unconstrained optimality

First-order necessary condition

Result – First-order criticality condition

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable and let x^* be an interior local minimum. Then

$$\nabla f(x^*) = 0.$$

Indeed, for every direction v ,

$$\phi(t) = f(x^* + tv)$$

has a local minimum at $t = 0$, hence

$$0 = \phi'(0) = \nabla f(x^*)^T v.$$

This holds for every v , so the gradient must vanish.

Warning

A stationary point need not be a minimum.

Result – Second-order classification

Around a stationary point x^* ,

$$f(x^* + h) = f(x^*) + \frac{1}{2}h^\top \nabla^2 f(x^*)h + o(\|h\|^2).$$

Hence:

- $\nabla^2 f(x^*) \succ 0 \Rightarrow$ strict local minimum;
- $\nabla^2 f(x^*) \prec 0 \Rightarrow$ strict local maximum;
- the Hessian is indefinite $\Rightarrow$ saddle point;
- a semidefinite Hessian may be inconclusive.

Reminder: Positivity of a Symmetric Matrix

Let $H = H^T$. Its quadratic form is $q_H(z) = z^T H z$.

Definitions

$H \succ 0 \iff z^T H z > 0$ for every $z \neq 0$, positive definite,

$H \succeq 0 \iff z^T H z \geq 0$ for every z , positive semidefinite,

$H \prec 0 \iff z^T H z < 0$ for every $z \neq 0$, negative definite.

If $z^T H z$ takes both positive and negative values, H is **indefinite**.

Easy Characterizations of Matrix Positivity

Eigenvalue test in any dimension

For symmetric H :

$$H \succ 0 \iff \lambda_i(H) > 0 \forall i, \quad H \succeq 0 \iff \lambda_i(H) \geq 0 \forall i.$$

Eigenvalues of both signs mean that H is indefinite.

Easy Characterizations of Matrix Positivity

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Eigenvalues of both signs mean that H is indefinite.

For $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, determinant tests are immediate:

$$\begin{aligned} H \succ 0 &\iff a > 0 \text{ and } ac - b^2 > 0, \\ H \prec 0 &\iff a < 0 \text{ and } ac - b^2 > 0, \\ H \text{ indefinite} &\iff ac - b^2 < 0. \end{aligned}$$

More generally, Sylvester's criterion uses the leading principal minors.

Example: Stationary Points in One Dimension

Consider

$$f(x) = x^3 - 3x^2 + 2.$$

Task. Find and classify all stationary points.

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$$f'(x) = 3x(x - 2),$$

so the candidates are $x = 0$ and $x = 2$.

Since $f''(x) = 6x - 6$,

$$f''(0) = -6 < 0 \Rightarrow \text{local maximum}, \quad f''(2) = 6 > 0 \Rightarrow \text{local minimum}.$$

Example: A Two-Dimensional Quadratic

Let

$$f(x, y) = x^2 + xy + y^2 - 4x - 2y.$$

The first-order condition gives

$$\nabla f(x, y) = \begin{pmatrix} 2x + y - 4 \\ x + 2y - 2 \end{pmatrix} = 0, \quad (x^*, y^*) = (2, 0).$$

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$$\nabla f(x, y) = \begin{pmatrix} 2x + y - 4 \\ x + 2y - 2 \end{pmatrix} = 0, \quad (x^*, y^*) = (2, 0).$$

Moreover,

$$H_f = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad 2 > 0, \quad \det(H_f) = 3 > 0.$$

Hence $H_f \succ 0$: $(2, 0)$ is the unique strict global minimum.

Example: Why Stationarity Is Not Enough

Consider $f(x, y) = x^2 - y^2$. We have $\nabla f(0, 0) = 0$, but

$$H_f = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \quad \det(H_f) = -4 < 0.$$

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Thus H_f is indefinite.

Equivalently,

$$f(t, 0) = t^2 > 0, \quad f(0, t) = -t^2 < 0$$

for $t \neq 0$. The function increases in one direction and decreases in another: $(0, 0)$ is a **saddle point**.

Worked Example: Classify with the Hessian

Classify the origin for

$$f_1(x, y) = x^2 + 4xy + 5y^2, \quad f_2(x, y) = x^2 - 4xy + y^2.$$

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$$H_1 = \begin{pmatrix} 2 & 4 \\ 4 & 10 \end{pmatrix}, \quad \det H_1 = 20 - 16 = 4 > 0, \quad (H_1)_{11} > 0.$$

Thus $H_1 \succ 0$: the origin is a strict minimum.

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Thus $H_1 \succ 0$: the origin is a strict minimum.

$$H_2 = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix}, \quad \det H_2 = 4 - 16 = -12 < 0.$$

Thus H_2 is indefinite: the origin is a saddle point.

Your Turn: Classify Two Quadratic Forms

Classify the origin for

$$g_1(x, y) = 3x^2 + 2xy + 2y^2, \quad g_2(x, y) = -x^2 - xy - y^2.$$

Task

Compute both Hessians and use the 2×2 determinant test.

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$$g_1(x, y) = 3x^2 + 2xy + 2y^2, \quad g_2(x, y) = -x^2 - xy - y^2.$$

Task

Compute both Hessians and use the 2×2 determinant test.

$$H_{g_1} = \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix}, \quad 6 > 0, \quad \det(H_{g_1}) = 20 > 0,$$

so the origin is a strict minimum of g_1 . Moreover,

$$H_{g_2} = \begin{pmatrix} -2 & -1 \\ -1 & -2 \end{pmatrix}, \quad -2 < 0, \quad \det(H_{g_2}) = 3 > 0,$$

so the origin is a strict maximum of g_2 .

When the Hessian says nothing

Compare

$$f(x) = x^4, \quad g(x) = -x^4.$$

Task. What does the second derivative test say at 0?

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Both functions satisfy

$$f'(0) = g'(0) = 0, \quad f''(0) = g''(0) = 0.$$

When the Hessian says nothing

Compare

$$f(x) = x^4, \quad g(x) = -x^4.$$

Task. What does the second derivative test say at 0?

Both functions satisfy

$$f'(0) = g'(0) = 0, \quad f''(0) = g''(0) = 0.$$

Yet 0 is a strict minimum of f and a strict maximum of g . We must inspect higher-order terms or use another argument.

Quadratic objectives

Let

$$f(x) = \frac{1}{2}x^T Qx + c^T x + r, \quad Q = Q^T.$$

Then

$$\nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q.$$

Your turn

Assume $Q \succ 0$. Find the minimizer.

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Then

$$\nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q.$$

Your turn

Assume $Q \succ 0$. Find the minimizer.

The first-order condition gives

$$Qx^* + c = 0 \implies \boxed{x^* = -Q^{-1}c.}$$

Positive definiteness makes this minimum unique and global.

Hands-on: least squares from first principles

Consider

$$f(x) = \frac{1}{2} \|Ax - b\|_2^2.$$

Task. Derive the normal equations.

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Expand:

$$f(x) = \frac{1}{2} (Ax - b)^T (Ax - b) = \frac{1}{2} x^T A^T A x - b^T A x + \frac{1}{2} b^T b.$$

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Therefore

$$\nabla f(x) = A^T (Ax - b), \quad \boxed{A^T A x = A^T b.}$$

If A has full column rank, $A^T A \succ 0$ and the solution is unique.

Section 3

Convexity: from local to global

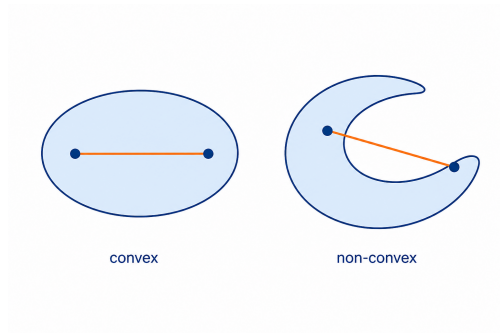
Convex sets and convex functions

A set $C \subset \mathbb{R}^d$ is convex if

$$x, y \in C, t \in [0, 1] \implies tx + (1 - t)y \in C.$$

A function $f : C \rightarrow \mathbb{R}$ is convex if

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y).$$



First-order characterization

Result – First-order characterization of convexity

If f is differentiable on a convex set, then

$$f \text{ convex} \iff f(y) \geq f(x) + \nabla f(x)^T(y - x)$$

for all x, y .

Global optimality consequence

If $\nabla f(x^*) = 0$, then

$$f(y) \geq f(x^*) \quad \forall y.$$

Every stationary point of a convex function is a global minimizer.

Second-order characterization

Result – Hessian characterization of convexity

If f is twice differentiable on an open convex set,

$$f \text{ convex} \iff \nabla^2 f(x) \succeq 0 \quad \forall x.$$

- $x \mapsto \frac{1}{2}x^T Qx + c^T x$ is convex iff $Q \succeq 0$.
- A nonnegative weighted sum of convex functions is convex.
- The pointwise maximum of convex functions is convex.

Worked Example: Prove Convexity Quickly

Let $x, a \in \mathbb{R}^d$, with a fixed. Determine whether the following functions are convex on their natural domains:

$$f(x) = e^{a^\top x}, \quad g(x) = \log(1 + e^x), \quad h(x, y) = x^2 - xy + y^2.$$

Worked Example: Prove Convexity Quickly

Let $x, a \in \mathbb{R}^d$, with a fixed. Determine whether the following functions are convex on their natural domains:

$$f(x) = e^{a^T x}, \quad g(x) = \log(1 + e^x), \quad h(x, y) = x^2 - xy + y^2.$$

$$\nabla f(x) = e^{a^T x} a, \quad \nabla^2 f(x) = e^{a^T x} a a^T \succeq 0, \quad g''(x) = \frac{e^x}{(1 + e^x)^2} > 0.$$

Indeed, for every $z \in \mathbb{R}^d$, $z^T \nabla^2 f(x) z = e^{a^T x} (a^T z)^2 \geq 0$.

Worked Example: Prove Convexity Quickly

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Indeed, for every $z \in \mathbb{R}^d$, $z^T \nabla^2 f(x) z = e^{a^T x} (a^T z)^2 \geq 0$.

$$\nabla^2 h = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

has eigenvalues 1 and 3. All three functions are convex.

Your Turn: Convex or Not?

Determine whether the following functions are convex on $\mathbb{R}^2$:

$$p(x, y) = x^4 + e^y, \quad q(x, y) = x^2 - y^2.$$

Task

Use the Hessian criterion and justify the conclusion globally.

Your Turn: Convex or Not?

Determine whether the following functions are convex on $\mathbb{R}^2$:

$$p(x, y) = x^4 + e^y, \quad q(x, y) = x^2 - y^2.$$

Task

Use the Hessian criterion and justify the conclusion globally.

$$\nabla^2 p(x, y) = \begin{pmatrix} 12x^2 & 0 \\ 0 & e^y \end{pmatrix} \succeq 0,$$

hence p is convex. In contrast,

$$\nabla^2 q = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

is indefinite, so q is not convex.

Strong convexity and smoothness

f is μ -strongly convex if

$$\nabla^2 f(x) \succeq \mu I.$$

Its gradient is L -Lipschitz if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|,$$

equivalently, for a twice differentiable f ,

$$\nabla^2 f(x) \preceq LI.$$

The ratio

$$\kappa = \frac{L}{\mu}$$

measures conditioning: large κ generally means slow first-order optimization.

Worked Example: Conditioning of a Quadratic

Let

$$f(x, y) = \frac{1}{2}(x^2 + 100y^2).$$

Task. Find μ , L , and κ .

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Worked Example: Conditioning of a Quadratic

Let

$$f(x, y) = \frac{1}{2}(x^2 + 100y^2).$$

Task. Find μ , L , and κ .

$$\nabla^2 f = \begin{pmatrix} 1 & 0 \\ 0 & 100 \end{pmatrix}.$$

Thus

$$\mu = 1, \quad L = 100, \quad \kappa = 100.$$

The level sets are elongated: one step size must handle two very different curvatures.

Your Turn: Read Conditioning from the Hessian

Let

$$g(x, y) = 2x^2 + 32y^2.$$

Task

Find the strong-convexity constant μ , smoothness constant L , and condition number $\kappa = L/\mu$.

Your Turn: Read Conditioning from the Hessian

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Task

Find the strong-convexity constant μ , smoothness constant L , and condition number $\kappa = L/\mu$.

$$\nabla^2 g = \begin{pmatrix} 4 & 0 \\ 0 & 64 \end{pmatrix}.$$

Therefore

$$\mu = 4, \quad L = 64, \quad \kappa = 16.$$

The slow direction is the x -axis and the steep direction is the y -axis.

Section 4

Algorithms for smooth optimization

Gradient descent

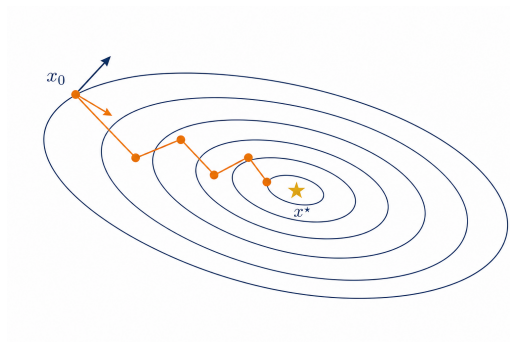
The gradient points in the direction of steepest local increase. Therefore

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$$

moves in a descent direction.

A practical implementation needs:

- an initial point x_0 ;
- a step-size rule α_k ;
- a stopping criterion, e.g. $\|\nabla f(x_k)\| \leq \varepsilon$;
- an iteration limit and numerical safeguards.



How small should the step be?

Result – Descent lemma and admissible step size

If ∇f is L -Lipschitz, then

$$f(x - \alpha \nabla f(x)) \leq f(x) - \alpha \left(1 - \frac{L\alpha}{2}\right) \|\nabla f(x)\|^2.$$

Therefore every fixed step

$$0 < \alpha < \frac{2}{L}$$

guarantees descent away from stationary points.

A standard conservative choice is $\alpha = 1/L$. If L is unknown, use backtracking line search.

Convergence under strong convexity

Result – Linear convergence rate

If f is μ -strongly convex and L -smooth, gradient descent with $\alpha = 1/L$ satisfies

$$f(x_k) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f(x^*)).$$

Interpretation

- The error decreases geometrically.
- A large condition number $\kappa = L/\mu$ slows convergence.
- Rescaling or preconditioning can change the geometry dramatically.

Newton's method

Replace f locally by its quadratic approximation:

$$f(x + p) \approx f(x) + \nabla f(x)^\top p + \frac{1}{2} p^\top \nabla^2 f(x) p.$$

Minimizing this model gives

$$\nabla^2 f(x_k) p_k = -\nabla f(x_k), \quad x_{k+1} = x_k + p_k.$$

Implementation rule

Solve the linear system; do not explicitly invert the Hessian.

Gradient descent or Newton?

	Gradient descent	Newton
Information	gradient	gradient + Hessian
Per iteration	relatively cheap	linear solve
Scaling	good for large d	demanding for large d
Local speed	linear	often quadratic
Robustness	step-size issue	Hessian may be indefinite

In practice: damping, line search, quasi-Newton methods, and automatic differentiation bridge the gap.

Projected gradient descent

Convex constrained problem and projection

We now want to solve the constrained convex problem

$$\min_{x \in C} f(x),$$

where f is differentiable and convex, and C is a nonempty closed convex set.

The Euclidean projection onto C is

$$\text{proj}_C(z) = \arg \min_{u \in C} \|u - z\|_2.$$

The projected-gradient update is

$$x_{k+1} = \text{proj}_C(x_k - \alpha_k \nabla f(x_k)).$$

Each iteration takes an unconstrained gradient step, then projects the result back onto the convex feasible set C .

Section 5

Equality constraints and Lagrange multipliers

Geometry of an equality constraint

Consider

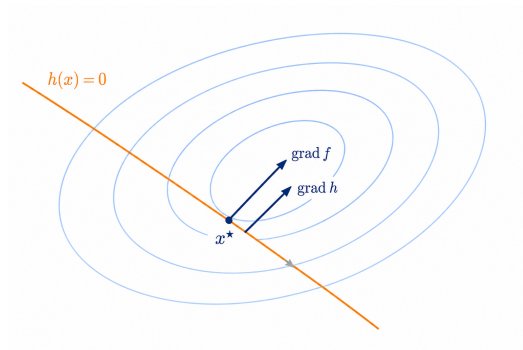
$$\min_x f(x) \quad \text{s.t.} \quad h(x) = 0.$$

At a regular feasible point, admissible first-order directions v satisfy

$$\nabla h(x)^T v = 0.$$

At a constrained optimum, $\nabla f(x^*)$ must also be orthogonal to every feasible direction. Therefore

$$\nabla f(x^*) + \lambda^* \nabla h(x^*) = 0.$$



The Lagrangian system

Result – Lagrange first-order conditions

Define

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T h(x).$$

Under a regularity condition, a constrained local optimum satisfies

$$\begin{aligned}\nabla_x \mathcal{L}(x^*, \lambda^*) &= 0, \\ h(x^*) &= 0.\end{aligned}$$

Necessary, not sufficient in general

These equations generate candidates. Without convexity or a second-order argument, a candidate may be a maximum or a saddle point.

Multipliers attach a price to infeasibility

Consider the equality-constrained primal problem

$$p^* = \min_x f(x) \quad \text{s.t.} \quad h(x) = 0, \quad \mathcal{L}(x, \lambda) = f(x) + \lambda^\top h(x).$$

For a fixed multiplier λ , the Lagrangian replaces the hard constraint by a penalty or reward proportional to its violation:

$$\lambda^\top h(x) = \sum_{i=1}^m \lambda_i h_i(x).$$

- x is the **primal variable**: it chooses the decision;
- λ is the **dual variable**: it prices constraint violations;
- a large $|\lambda_i|$ indicates that constraint i strongly affects value.

Sensitivity interpretation

Under regularity, the optimal multiplier measures the marginal change in the optimal value when the corresponding constraint is perturbed.

The dual function gives one bound per multiplier

For a fixed λ , minimize the Lagrangian over all primal decisions:

$$q(\lambda) = \inf_x \mathcal{L}(x, \lambda).$$

The function q is concave in λ , even when the primal problem is not convex, because it is a pointwise infimum of affine functions of λ .

If x is feasible, then $h(x) = 0$, so

$$q(\lambda) \leq \mathcal{L}(x, \lambda) = f(x).$$

From a family of bounds to the dual problem

$$d^* = \sup_{\lambda} q(\lambda).$$

Each λ produces a lower bound; the dual chooses the tightest one.

Weak duality leaves a measurable gap

Taking the best dual bound and the best feasible primal value gives

Result – Weak duality

$$\underbrace{d^*}_{\sup_{\lambda} q(\lambda)} \leq \underbrace{p^*}_{\inf_{h(x)=0} f(x)}.$$

This is **weak duality**; it does not require convexity.

For any primal-feasible x and any multiplier λ ,

$$\underbrace{f(x) - q(\lambda)}_{\text{primal-dual gap}} \geq 0.$$

Why the gap is useful

A small gap certifies that the primal candidate and the dual bound are close to optimal. Algorithms often monitor this gap as a stopping criterion.

Strong duality means that the best bound is exact

Result – Strong duality

Strong duality means that the primal–dual gap vanishes at the optimum:

$$d^* = p^*.$$

A standard convex setting is:

- f and the inequality functions are convex;
- equality constraints are affine;
- a constraint qualification holds, commonly Slater's condition.

Convexity alone is not the whole story

Without an appropriate regularity condition, a positive duality gap may remain or the dual optimum may fail to be attained.

Under strong duality, solving the dual problem can recover the primal value, certify optimality, and sometimes be computationally easier.

Strong duality turns optimization into a saddle problem

The primal formulation minimizes after enforcing feasibility:

$$p^* = \min_x \max_{\lambda} \mathcal{L}(x, \lambda).$$

The dual formulation maximizes the best lower bound:

$$d^* = \max_{\lambda} \min_x \mathcal{L}(x, \lambda).$$

When strong duality holds and optimal solutions exist,

$$\min_x \max_{\lambda} \mathcal{L}(x, \lambda) = \max_{\lambda} \min_x \mathcal{L}(x, \lambda).$$

The primal–dual solution (x^*, λ^*) is a saddle point:

$$\mathcal{L}(x^*, \lambda) \leq \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*).$$

Moving in x can only increase the Lagrangian away from the minimizer, while moving in λ can only decrease it away from the maximizer.

Gradient descent–ascent is a primal–dual algorithm

For a differentiable Lagrangian, a direct saddle-point iteration is

$$\begin{aligned}x_{k+1} &= x_k - \alpha_k \nabla_x \mathcal{L}(x_k, \lambda_k), & \text{gradient descent in } x, \\ \lambda_{k+1} &= \lambda_k + \beta_k \nabla_\lambda \mathcal{L}(x_k, \lambda_k), & \text{gradient ascent in } \lambda.\end{aligned}$$

Since $\nabla_\lambda \mathcal{L}(x, \lambda) = h(x)$,

$$\lambda_{k+1} = \lambda_k + \beta_k h(x_k).$$

- the primal step reduces the objective for the current multiplier;
- the dual step increases the price of violated constraints;
- step sizes and convex–concave structure determine convergence.

Practical warning

Simultaneous descent–ascent can oscillate. Damping, extragradient methods, or an augmented Lagrangian often improve stability.

Worked Example: Minimum Distance to a Line

Solve

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y = 1.$$

Worked Example: Minimum Distance to a Line

Solve

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y = 1.$$

$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1).$$

The first-order conditions are

$$2x + \lambda = 0, \quad 2y + \lambda = 0, \quad x + y = 1.$$

Worked Example: Minimum Distance to a Line

Solve

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad x + y = 1.$$

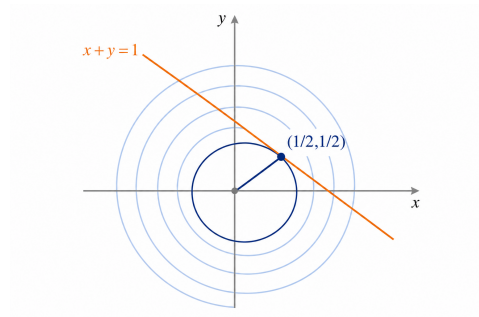
$$\mathcal{L}(x, y, \lambda) = x^2 + y^2 + \lambda(x + y - 1).$$

The first-order conditions are

$$2x + \lambda = 0, \quad 2y + \lambda = 0, \quad x + y = 1.$$

Thus $x = y$ and

$$x^* = y^* = \frac{1}{2}, \quad \lambda^* = -1.$$



Your Turn: Minimum Distance to Another Line

Solve

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad 2x - y = 3.$$

Task

Write the Lagrangian and solve the three first-order equations.

Your Turn: Minimum Distance to Another Line

Solve

$$\min_{x,y} x^2 + y^2 \quad \text{s.t.} \quad 2x - y = 3.$$

Task

Write the Lagrangian and solve the three first-order equations.

$$\mathcal{L} = x^2 + y^2 + \lambda(2x - y - 3),$$

so $2x + 2\lambda = 0$, $2y - \lambda = 0$, and $2x - y = 3$. Therefore

$$\lambda^* = -\frac{6}{5}, \quad (x^*, y^*) = \left(\frac{6}{5}, -\frac{3}{5}\right).$$

Several equality constraints

For $h : \mathbb{R}^d \rightarrow \mathbb{R}^m$,

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^\top h(x).$$

The stationarity condition is

$$\nabla f(x^*) + Dh(x^*)^\top \lambda^* = 0.$$

LICQ: the condition to check

$\nabla h_1(x^*), \dots, \nabla h_m(x^*)$ are linearly independent.

Without regularity, Lagrange multipliers may fail to describe an optimum.

Worked Example: Why Regularity Matters

Consider

$$\min_x x \quad \text{s.t.} \quad x^2 = 0.$$

The only feasible point is $x^* = 0$, so it is the optimizer.

Task. Can the Lagrange stationarity equation hold?

Worked Example: Why Regularity Matters

Consider

$$\min_x x \quad \text{s.t.} \quad x^2 = 0.$$

The only feasible point is $x^* = 0$, so it is the optimizer.

Task. Can the Lagrange stationarity equation hold?

$$\mathcal{L}(x, \lambda) = x + \lambda x^2, \quad \partial_x \mathcal{L}(0, \lambda) = 1.$$

Worked Example: Why Regularity Matters

Consider

$$\min_x x \quad \text{s.t.} \quad x^2 = 0.$$

The only feasible point is $x^* = 0$, so it is the optimizer.

Task. Can the Lagrange stationarity equation hold?

$$\mathcal{L}(x, \lambda) = x + \lambda x^2, \quad \partial_x \mathcal{L}(0, \lambda) = 1.$$

No multiplier can make the derivative vanish. Here $h'(0) = 0$, so the constraint gradient is not regular.

Your Turn: Detect Another Failure of Regularity

Consider

$$\min_{x,y} x + y \quad \text{s.t.} \quad x^2 + y^2 = 0.$$

Task

Identify the optimizer and test the Lagrange stationarity equation.

Your Turn: Detect Another Failure of Regularity

Consider

$$\min_{x,y} x + y \quad \text{s.t.} \quad x^2 + y^2 = 0.$$

Task

Identify the optimizer and test the Lagrange stationarity equation.

The feasible set contains only $(0,0)$, so it is the optimizer. However,

$$\nabla f(0,0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \nabla h(0,0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

No λ can satisfy $\nabla f + \lambda \nabla h = 0$. The constraint gradient vanishes, so the regularity condition fails.

Application: fully invested minimum variance

Solve

$$\min_w \frac{1}{2} w^T \Sigma w \quad \text{s.t.} \quad \mathbf{1}^T w = 1,$$

Here $\Sigma \succ 0$ and $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^d$ is the **vector of ones**.

Task. Derive the optimal weights.

Application: fully invested minimum variance

Solve

$$\min_w \frac{1}{2} w^T \Sigma w \quad \text{s.t.} \quad \mathbf{1}^T w = 1,$$

Here $\Sigma \succ 0$ and $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^d$ is the **vector of ones**.

Task. Derive the optimal weights.

$$\Sigma w + \lambda \mathbf{1} = 0 \quad \implies \quad w = -\lambda \Sigma^{-1} \mathbf{1}.$$

Enforcing the budget constraint gives

$$-\lambda = \frac{1}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}.$$

Application: fully invested minimum variance

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Enforcing the budget constraint gives

$$-\lambda = \frac{1}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}.$$

$$w^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}.$$

Section 6

Inequality constraints and KKT conditions

KKT conditions

Consider

$$\min_x f(x) \quad \text{s.t.} \quad h(x) = 0, \quad g(x) \leq 0,$$

with

$$\mathcal{L}(x, \nu, \lambda) = f(x) + \nu^T h(x) + \lambda^T g(x).$$

With the convention $g(x) \leq 0$, the inequality multiplier must satisfy $\lambda \geq 0$; the equality multiplier ν is unrestricted.

Result – KKT necessary conditions

Under suitable regularity, a local optimum satisfies:

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \nu^*, \lambda^*) &= 0 && \text{stationarity,} \\ h(x^*) &= 0, \quad g(x^*) \leq 0 && \text{primal feasibility,} \\ \lambda^* &\geq 0 && \text{dual feasibility,} \\ \lambda_i^* g_i(x^*) &= 0 && \text{complementary slackness.} \end{aligned}$$

Active or inactive?

For one inequality $g(x) \leq 0$:

- if $g(x^*) < 0$, the constraint is inactive and $\lambda^* = 0$;
- if $\lambda^* > 0$, complementary slackness forces $g(x^*) = 0$;
- an active constraint can still have $\lambda^* = 0$.

Practical method

Guess the active set, solve the corresponding equations, then verify all signs and inequalities.

Hands-on: an inactive constraint

Solve

$$\min_x (x - 2)^2 \quad \text{s.t.} \quad x \geq 1.$$

Write $g(x) = 1 - x \leq 0$.

Hands-on: an inactive constraint

Solve

$$\min_x (x - 2)^2 \quad \text{s.t.} \quad x \geq 1.$$

Write $g(x) = 1 - x \leq 0$.

$$\mathcal{L}(x, \lambda) = (x - 2)^2 + \lambda(1 - x),$$

and stationarity gives

$$2(x - 2) - \lambda = 0.$$

Hands-on: an inactive constraint

Solve

$$\min_x (x - 2)^2 \quad \text{s.t.} \quad x \geq 1.$$

Write $g(x) = 1 - x \leq 0$.

$$\mathcal{L}(x, \lambda) = (x - 2)^2 + \lambda(1 - x),$$

and stationarity gives

$$2(x - 2) - \lambda = 0.$$

The unconstrained minimizer $x^* = 2$ is feasible and strictly inside the set. Hence

$x^* = 2, \quad \lambda^* = 0.$

Hands-on: a binding constraint

Solve

$$\min_x x^2 \quad \text{s.t.} \quad x \geq 1.$$

Hands-on: a binding constraint

Solve

$$\min_x x^2 \quad \text{s.t.} \quad x \geq 1.$$

With $g(x) = 1 - x \leq 0$,

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(1 - x), \quad 2x - \lambda = 0.$$

Hands-on: a binding constraint

Solve

$$\min_x x^2 \quad \text{s.t.} \quad x \geq 1.$$

With $g(x) = 1 - x \leq 0$,

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(1 - x), \quad 2x - \lambda = 0.$$

The constraint must bind, so $x^* = 1$. Stationarity then gives

$$x^* = 1, \quad \lambda^* = 2.$$

All four KKT conditions are satisfied.

Worked Example: A Two-Dimensional Boundary

Solve

$$\min_{x,y} (x-1)^2 + (y-2)^2 \quad \text{s.t.} \quad x+y \leq 1.$$

Worked Example: A Two-Dimensional Boundary

Solve

$$\min_{x,y} (x-1)^2 + (y-2)^2 \quad \text{s.t.} \quad x+y \leq 1.$$

With $g(x,y) = x+y-1$,

$$2(x-1) + \lambda = 0, \quad 2(y-2) + \lambda = 0.$$

Hence $x = 1 - \lambda/2$ and $y = 2 - \lambda/2$.

Worked Example: A Two-Dimensional Boundary

Solve

$$\min_{x,y} (x-1)^2 + (y-2)^2 \quad \text{s.t.} \quad x+y \leq 1.$$

With $g(x,y) = x+y-1$,

$$2(x-1) + \lambda = 0, \quad 2(y-2) + \lambda = 0.$$

Hence $x = 1 - \lambda/2$ and $y = 2 - \lambda/2$.

The unconstrained point $(1, 2)$ is infeasible, so the constraint binds:

$$x + y = 3 - \lambda = 1.$$

Thus

$$\lambda^* = 2, \quad (x^*, y^*) = (0, 1).$$

Your Turn: Project onto a Half-Space

Solve

$$\min_{x,y} (x-2)^2 + (y+1)^2 \quad \text{s.t.} \quad x - y \leq 1.$$

Task

Determine whether the constraint binds, then solve the KKT equations.

Your Turn: Project onto a Half-Space

Solve

$$\min_{x,y} (x-2)^2 + (y+1)^2 \quad \text{s.t.} \quad x - y \leq 1.$$

Task

Determine whether the constraint binds, then solve the KKT equations.

The unconstrained point $(2, -1)$ violates $x - y \leq 1$. With $g(x, y) = x - y - 1$,

$$x = 2 - \frac{\lambda}{2}, \quad y = -1 + \frac{\lambda}{2}.$$

Binding feasibility gives $3 - \lambda = 1$, hence

$$\lambda^* = 2, \quad (x^*, y^*) = (1, 0).$$

When KKT is sufficient

Result – Convex KKT sufficiency

Suppose:

- f and every g_i are convex;
- every equality constraint is affine;
- a constraint qualification holds.

Then any point satisfying the KKT conditions is a **global optimum**.

Nonconvex problems

KKT conditions are generally only necessary local conditions. Candidate points still need to be compared or classified.

Section 7

Portfolio optimization

Returns and risk become an optimization problem

For d assets, let $R = (R_1, \dots, R_d)^T \in \mathbb{R}^d$ be the **random vector of asset returns** over one investment period, and let

$$w \in \mathbb{R}^d, \quad \mu = \mathbb{E}[R], \quad \Sigma = \text{Cov}(R).$$

Then

$$\mathbb{E}[w^T R] = \mu^T w, \quad \text{Var}(w^T R) = w^T \Sigma w.$$

Typical portfolio constraints are

$$\mathbf{1}^T w = 1, \quad w \geq 0, \quad \ell \leq w \leq u.$$

When $\Sigma \succeq 0$, variance is a convex quadratic objective.

Mean–variance optimization balances two objectives

A standard formulation is

$$\min_w \frac{1}{2} w^T \Sigma w - \gamma \mu^T w \quad \text{s.t.} \quad \mathbf{1}^T w = 1.$$

Here $\gamma > 0$ controls the reward assigned to expected return.

Derive the first-order system

Write the Lagrangian and solve for w in terms of the multiplier.

Mean–variance optimization balances two objectives

A standard formulation is

$$\min_w \frac{1}{2} w^T \Sigma w - \gamma \mu^T w \quad \text{s.t.} \quad \mathbf{1}^T w = 1.$$

Here $\gamma > 0$ controls the reward assigned to expected return.

Derive the first-order system

Write the Lagrangian and solve for w in terms of the multiplier.

$$\Sigma w - \gamma \mu + \lambda \mathbf{1} = 0, \quad \mathbf{1}^T w = 1.$$

Thus

$$w = \Sigma^{-1}(\gamma \mu - \lambda \mathbf{1}),$$

with λ determined by the budget constraint.

A target return traces the efficient frontier

Fix a target return m . The minimum-risk portfolio solves

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad \mu^\top w = m.$$

Define

$$C = \begin{pmatrix} \mathbf{1}^\top \\ \mu^\top \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ m \end{pmatrix}.$$

A target return traces the efficient frontier

Fix a target return m . The minimum-risk portfolio solves

$$\min_w \frac{1}{2} w^T \Sigma w \quad \text{s.t.} \quad \mathbf{1}^T w = 1, \quad \mu^T w = m.$$

Define

$$C = \begin{pmatrix} \mathbf{1}^T \\ \mu^T \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ m \end{pmatrix}.$$

The equality-constrained quadratic formula gives

$$w^* = \Sigma^{-1} C^T (C \Sigma^{-1} C^T)^{-1} d.$$

Varying m generates the efficient frontier in mean–variance space.

Long-only portfolios use KKT conditions

Add the constraints $w_i \geq 0$, written as $-w_i \leq 0$:

$$\min_w \frac{1}{2} w^T \Sigma w - \gamma \mu^T w \quad \text{s.t.} \quad \mathbf{1}^T w = 1, \quad -w \leq 0.$$

The KKT stationarity equation is

$$\Sigma w - \gamma \mu + \lambda \mathbf{1} - \eta = 0,$$

with

$$w \geq 0, \quad \eta \geq 0, \quad \eta_i w_i = 0.$$

Since the problem is convex, KKT conditions are sufficient under the usual constraint qualification.

Real portfolios add operational constraints

Practical models commonly include

- position bounds $\ell_i \leq w_i \leq u_i$;
- sector or factor-exposure limits;
- turnover control $\|w - w^{\text{old}}\|_1 \leq \tau$;
- transaction costs and tracking error;
- stress constraints across several scenarios.

Modeling principle

Preserve convexity when possible: quadratic risk with linear constraints is a quadratic program that can be solved reliably at realistic scale.

Hands-on synthesis: constrained rebalancing

Consider

$$\min_w \frac{1}{2} w^T \Sigma w - \gamma \mu^T w$$

subject to

$$\mathbf{1}^T w = 1, \quad w \geq 0, \quad \|w - w^{\text{old}}\|_2^2 \leq \tau^2.$$

Task. Write the KKT conditions and identify the appropriate solver.

Hands-on synthesis: constrained rebalancing

Consider

$$\min_w \frac{1}{2} w^T \Sigma w - \gamma \mu^T w$$

subject to

$$\mathbf{1}^T w = 1, \quad w \geq 0, \quad \|w - w^{\text{old}}\|_2^2 \leq \tau^2.$$

Task. Write the KKT conditions and identify the appropriate solver.

Introduce $\lambda \in \mathbb{R}$, $\eta \geq 0$ for $-w \leq 0$, and $\rho \geq 0$ for $\|w - w^{\text{old}}\|_2^2 - \tau^2 \leq 0$. Then

$$\Sigma w - \gamma \mu + \lambda \mathbf{1} - \eta + 2\rho(w - w^{\text{old}}) = 0.$$

Together with primal and dual feasibility, complementary slackness is

$$\eta_i w_i = 0 \quad \forall i, \quad \rho(\|w - w^{\text{old}}\|_2^2 - \tau^2) = 0.$$

When $\Sigma \succeq 0$, this is a convex quadratically constrained quadratic program; under the usual qualification, KKT is sufficient and a convex QCQP solver returns the global optimum.

Section 8

Putting the toolkit to work

A robust problem-solving checklist

- 1 Write the decision variable, objective, and feasible set.
- 2 Check dimensions, differentiability, and convexity.
- 3 Compute first-order conditions.
- 4 Use Hessian information or convexity to classify candidates.
- 5 For constraints, state a sign convention and write all KKT conditions.
- 6 Choose an algorithm that respects the structure.
- 7 Verify feasibility, stationarity, and numerical convergence.
- 8 Interpret the result in the original application.

Final challenge

Consider the ridge-regression problem

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \frac{\lambda}{2} \|x\|_2^2, \quad \lambda > 0.$$

Tasks.

- 1 Prove that the objective is strongly convex.
- 2 Derive the unique minimizer.
- 3 Write one gradient-descent iteration.

Final challenge

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Tasks.

- 1 Prove that the objective is strongly convex.
- 2 Derive the unique minimizer.
- 3 Write one gradient-descent iteration.

$$\nabla f(x) = A^T(Ax - b) + \lambda x, \quad \nabla^2 f(x) = A^T A + \lambda I \succeq \lambda I.$$

Final challenge

Consider the ridge-regression problem

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \frac{\lambda}{2} \|x\|_2^2, \quad \lambda > 0.$$

Tasks.

- 1 Prove that the objective is strongly convex.
- 2 Derive the unique minimizer.
- 3 Write one gradient-descent iteration.

$$\nabla f(x) = A^T(Ax - b) + \lambda x, \quad \nabla^2 f(x) = A^T A + \lambda I \succeq \lambda I.$$

$$\boxed{x^* = (A^T A + \lambda I)^{-1} A^T b,} \quad x_{k+1} = x_k - \alpha(A^T(Ax_k - b) + \lambda x_k).$$

Model choice: what structure do we see?

Structure	Natural first method
smooth, unconstrained, large scale	gradient / quasi-Newton
smooth, accurate local solve	Newton / damped Newton
simple convex set	projected gradient
equality constraints	Lagrange system
convex inequalities	KKT / convex solver
long-only portfolio with bounds	quadratic-programming solver
quadratic objective and linear constraints	QP solver

Before choosing an algorithm, identify convexity, smoothness, constraints, and problem dimension.